05 Polymers and Proteins Problem-solving Approach

Basis: from the 2016–2025 Chemistry ATAR past exam questions in Section Two and Section Three, the sub-questions tagged PRO, 7 patterns in total, with 89 tag occurrences altogether (consistent with INDEX.md; excludes Section One multiple-choice questions). Marking points are taken from the key in the marking rubric, paraphrased, not copied verbatim; the original papers and marking files are kept in a local ignored directory. Exam question numbering format: `2021E-Q34(g)` = 2021 Section Three Question 34 (g); `2022S-Q29(ai)` = 2022 Section Two Question 29 (a)(i). Amino acid side-chain classification is based on the structure table of the 20 α-amino acids in the 2024 Data Booklet, and the key for 2022S-Q29(e). Pattern definitions are taken from `tools/catalogue_CHE.json`; labels are draft (draft_for_review); this file has not been independently reviewed. ★ markers: ranked by total score for each pattern (INDEX.md), taking roughly the top quarter; this file covers PRO6 and PRO4.

Marking Panel Reminders

- Protein structure is the backbone of PRO: PRO6 has 25 tags, making it the most heavily tagged pattern in this topic. 2024E-Q37 uses the structure of lipase B as its context, connecting several sub-questions on secondary structure, hydrogen bonds and disulfide bonds.
- Polymers and proteins questions performed relatively well in 2021: the 2021 report states that the three best-performing questions in Section Three were Question 34 (polymers and proteins), Question 35 (titration calculations) and Question 36 (empirical formula).
- 2020 condensation polymer question: 2020S-Q28 (poly(ethylene adipate), a biodegradable polymer) averaged 68.45%.
- Structure-drawing questions must include all H atoms: most drawing questions' keys have a clear rule of "one minor error deducts 1 mark"; drawing incorrectly or omitting one H loses marks.
- The report does not provide separate comments for each PRO pattern; the report entries in the following sections cite only content mentioned in the original report.

Pattern 1: Addition polymers: monomer–polymer interconversion

Identification: the question stem gives a monomer (a small molecule containing C=C) and asks to "draw three repeating units" or "draw and name the polymer"; or gives the polymer structure and asks to "draw and name the monomer"; commonly seen with polyacrylic acid, PMMA, synthetic nail polymers, etc.

Steps

From monomer to polymer

1. Identify the C=C in the monomer and open the double bond: each of the two carbons forms a new σ bond and connects to a carbon in the adjacent monomer, so the backbone is made up entirely of C–C single bonds.
2. Keep the substituents (side groups) on the two carbons attached to their original carbons; do not move them.
3. Draw at least three repeating units, shown with brackets and subscript n; do not add end groups at either end (use broken lines or keep the chain extending).
4. Write the polymer name, consistent with the structure.

From polymer to monomer

1. On the backbone, identify two adjacent carbons; the single bond between them must be restored to a double bond.
2. Keep the side groups as substituents of the monomer.
3. Name the monomer as an alkene, and write its name and structural formula.

Marking points (typical wording in the key): the monomer contains C=C and its name is consistent with the structure (2016S-Q29(a), 1 mark each); the polymer contains at least three repeating units, with the name consistent with the structure (2016S-Q29(b)); three monomers are correctly connected, with the ends not terminated (2022S-Q31(a)); addition polymerisation characteristics (2019E-Q38(e): one monomer, one

species added to each end of the double bond).

Common errors (exam report)

- Fewer than three repeating units, or drawing only the monomer rather than the polymer.
- Side groups attached to the wrong carbon. In the PMMA repeating unit, the methyl group and the ester group are attached to the same backbone carbon and must not be separated (the key for 2019E-Q38(c) requires a continuous backbone, no end groups, and the correct repeating unit).
- Adding end groups: the key explicitly requires "ends are not terminated" (2022S-Q31(a)).
- The key for 2020S-Q28(c) writes addition (polymerisation/reaction); answering condensation scores no marks.

Past exam questions: 2016S-Q29(a), 2016S-Q29(b), 2017E-Q36(e), 2019E-Q38(c), 2019E-Q38(e), 2020S-Q28(c), 2021E-Q34(f), 2021E-Q34(g), 2021E-Q34(h), 2022S-Q31(a)

Pattern 2: Condensation polymers: polyesters and polyamides

Identification: the question stem contains "condensation polymer", "polyester", "polyamide", "Kevlar", "PET", "by-product" or "the two monomers"; or asks you to draw the polymer, label the ester bond or amide bond, or write the small-molecule by-product.

Steps

From monomer to polymer

1. Determine the reactive groups of the monomers: condensation monomers generally have two reactive functional groups, such as –COOH and –OH (forming an ester bond –C(=O)–O–), or –COOH and $–NH_2$ (forming an amide bond –C(=O)–NH–).
2. Form an ester bond or amide bond between adjacent monomers, while eliminating a small molecule (H_2O, or HCl, etc.).
3. Draw at least two repeating units (the key usually requires two or three).
4. Label the ester bond or amide bond, and write the by-product.

From polymer to monomer

1. Locate C(=O)–O (ester bond) or C(=O)–N (amide bond).
2. Break that bond: add –OH to the carbonyl carbon end, and add H to the oxygen or nitrogen end.
3. This gives two monomers (such as a diacid and a diol, or a diacid and a diamine); draw their structures separately.

Marking points (typical wording in the key): both monomer structures are correct (2017E-Q36(c): terephthalic acid and ethylene glycol, 2 marks each; only minor errors score 1 mark; 2016S-Q29(d), 2022S-Q31(c): the diamine and dicarboxylic acid for Kevlar); the repeating unit structure is correct, with the ester bond labelled (2018S-Q30(a), 2020S-Q28(a): "one minor error" deducts 1 mark); the by-product is written as water or H_2O (2018S-Q30(b)); the two characteristics of condensation polymerisation (2024S-Q35(b): the monomer has two reactive groups; a small molecule is also produced).

Common errors (exam report)

- Writing the ester bond as C–O instead of C(=O)–O, or omitting the ester bond, is treated as a major error in the key, and that part scores 0 marks (2018S-Q30(a), 2020S-Q28(a)); only minor errors such as missing H or terminated end groups deduct 1 mark.
- 2021E-Q34(g), (h) require distinguishing addition polymerisation from condensation polymerisation: condensation produces a small molecule, whereas addition does not; for (h), any two points are sufficient.
- 2023E-Q36(d) requires alanine and lactic acid each to form three repeating units: condensation of alanine forms amide bonds (peptide bonds), and condensation of lactic acid forms ester bonds.
- Report entry for 2020S-Q28: poly(ethylene adipate) is a biodegradable polyester, average 68.45%.
- 2025S-Q28(a): the monomer of P3HB; the key diagram is 3-hydroxybutanoic acid HO–CH(CH_3)–CH_2–

COOH, 2 marks in total; minor errors such as wrong methyl position, incorrect number of H, or improper connection score only 1 mark.

Past exam questions: 2016S-Q29(d), 2017E-Q36(c), 2017E-Q36(e), 2018S-Q30(a), 2018S-Q30(b), 2019E-Q38(e), 2020S-Q28(a), 2020S-Q28(bii), 2021E-Q34(g), 2021E-Q34(h), 2022S-Q31(c), 2023E-Q36(d), 2024S-Q35(b), 2025S-Q28(a), 2025S-Q28(b)

Pattern 3: Plastic structure, properties and uses

Identification: the question stem asks to "state one use … making reference to its property", "state one common use and two properties", "which polymer is more appropriate … justify", or "predict the degree of rigidity".

Steps

1. Determine the polymer type: addition or condensation; linear, branched or cross-linked (cross-linked thermosetting polymers are rigid and do not soften on heating; linear or branched thermoplastic polymers can soften on heating).
2. Explain properties from structure: intermolecular forces (e.g. interchain hydrogen bonds in Kevlar give high strength; dipole–dipole forces in PET), chain length and branching (affect crystallinity and rigidity).
3. Match each use to one property: pair one use with a related property; do not write a property unrelated to the use.
4. When comparing two polymers, give a comparison of two properties for each, and state which is more suitable.
5. Conclusion sentence pattern: "X is more appropriate because it is more … than Y".

Marking points (typical wording in the key): 1 mark for the use and 1 mark for the property related to the use (2016S-Q29(c)); for PET, the use and property are each scored; accepted uses include packaging, bottles, fibres, magnetic tape, etc., and properties include gas barrier, moisture resistance, transparency, high mechanical strength, chemical resistance, etc. (2017E-Q36(d)); in the comparison choice question, each choice is worth 1–2 marks (4 marks in total), including the choice conclusion and a comparison of two properties (2019E-Q38(a)); P3HB is linear and has only methyl side chains, so it is predicted to have higher rigidity (2025S-Q28(c)).

Common errors (exam report)

- Property and use do not correspond: e.g. "strong" is not linked to "impact resistance of safety glasses".
- Writing only the property name without explaining its relationship to structure (chains, branching, forces).
- This pattern has 4 parts; there are no separate comments in the report, so evidence is limited.

Past exam questions: 2016S-Q29(c), 2017E-Q36(d), 2019E-Q38(a), 2025S-Q28(c)

Pattern 4: α-Amino Acids and Zwitterions ★

Identifying features: the question asks you to "draw the structure at low pH / high pH", "draw as a zwitterion", "identify an amino acid with a non-polar side chain", "name the three amino acids", "explain why … pH dependent"; or requires judgement using structures from the data booklet.

Steps

1. General formula: $H_2N–CH(R)–COOH$; the α-carbon is bonded to the amino group, carboxyl group, hydrogen and side chain R.
2. Draw the form present according to pH:
    - Low pH (acidic): the amino group is protonated to $–NH_3^+$, the carboxyl group remains –COOH, overall charge +1;

- Neutral (pH ≈ 7, near the isoelectric point): zwitterion, $–NH_3^+$ and $–COO^–$ are both present, overall charge 0;
- High pH (basic): the carboxyl group is deprotonated to $–COO^–$, the amino group remains $–NH_2$, overall charge −1.

3. Formation of a zwitterion: the carboxyl group (the stronger acid) donates H^+ to the amino group (base), forming the same molecule with positive and negative charges.
4. Side-chain classification (check the data booklet):

    - Non-polar: Ala, Gly, Val, Leu, Ile, Met, Phe, Pro (the 8 listed in the key for 2022S-Q29(e) are alanine, glycine, isoleucine, leucine, methionine, phenylalanine, proline, valine);
    - Polar uncharged: Ser, Thr, Cys, Tyr, Asn, Gln;
    - Acidic: Asp, Glu; basic: Lys, Arg, His.
    - The side chain of tryptophan (Trp) contains an indole ring and N–H; the 2022 key does not list it as non-polar, so when answering, use the structure in the data booklet and give a reason.

Marking points (typical wording in the key): in 2018E-Q39(c), the low-pH and high-pH structures are worth 2 marks each (the question requires all H atoms to be shown; minor errors incur a 1-mark deduction each); 2020E-Q35(f) is worth 2 marks in total, 1 mark for the acidic structure and 1 mark for the basic structure, and an incorrectly drawn amino acid scores at most 1 mark; explaining pH dependence: at neutral pH it exists as a zwitterion; at low pH the amino group is protonated; at high pH the carboxyl group is deprotonated (2018E-Q39(d), 3 marks); definition and formation of a zwitterion (2024S-Q29(a): the same molecule carries both positive and negative charges; the carboxyl group donates H^+ to the amino group; correctly drawing the serine zwitterion; 4 marks in total); a zwitterion exists only at neutral pH or pH ≈ 7 (2022S-Q29(aii)); amino acids with non-polar side chains (2022S-Q29(e)).

Common errors (examiner reports)

- Drawing NH_2 and $COO^–$ together but mistaking it for a zwitterion (that is the high-pH form, not a zwitterion).
- Omitting COOH at low pH, or drawing the carboxyl group as $COO^–$.
- Omitting H when drawing structures: the question for 2018E-Q39(c) requires "show all hydrogen atoms"; the question for 2020E-Q35(f) does not have this requirement.
- The reports do not give a separate comment on this pattern. 2023E-Q36(c) and 2024S-Q29(b) use ionic attraction between zwitterions to explain high melting points; see ORG4 Pattern 4.

Past exam questions: 2016S-Q36(aii), 2017E-Q40(a), 2017E-Q40(e), 2018E-Q39(c), 2018E-Q39(d), 2019E-Q41(a), 2020E-Q35(f), 2021E-Q34(a), 2022S-Q29(ai), 2022S-Q29(aii), 2022S-Q29(e), 2023E-Q36(c), 2024S-Q29(a), 2024S-Q29(b)

Pattern 5: Peptide Bonds and Polypeptides

Identifying features: the question asks you to "circle the peptide bonds", "draw a section of the polypeptide / dipeptide / tripeptide", "list the amino acids in order", "write an equation for the reaction between two amino acids".

Steps

1. Peptide bond: C(=O)–NH, linking the carboxyl carbon of one amino acid to the amino nitrogen of the next amino acid. When circling it, include C=O, N, and the H on N.
2. Formation: the –COOH of one amino acid condenses with the $–NH_2$ of another, with loss of H_2O, to give a peptide bond. When writing the equation, use condensed structures and show water as a product.
3. Drawing a polypeptide: write the N-terminus on the left (H_2N–…) and the C-terminus on the right (…–COOH), with each amino acid's R group attached to the α-carbon; arrange them in the order given in the question.
4. Reading the sequence: read from the N-terminus; write amino acids in full names, not abbreviations

(explicitly required by 2019E-Q41(a)).

5. Link to Pattern 6: the primary structure is this sequence.

Marking points (typical wording in the key): both peptide bonds correctly circled score 2 marks; circling only one scores 1 mark (2016S-Q36(ai)); for 2019E-Q41(b), circling one peptide bond scores 1 mark; correct amino acid order (2019E-Q41(a): all 5 correct score 2 marks; at least 3 in the correct order score 1 mark; writing the full names scores an additional 1 mark); in a complete structure, correct peptide bonds and order score 3 marks, while capped termini or incorrect order but correct amino acid structures score 2 marks (2018E-Q39(a)); equation for a dipeptide (2020E-Q35(e): condensation of phenylalanine and leucine).

Common errors (examiner reports)

- Circling the peptide bond as only a C–N single bond, without C=O.
- Reading the sequence from the C-terminus, reversing the sequence.
- For 2025E-Q36(d), when drawing three residues, the key requires correct R groups, correct order and uncapped termini at both ends; missing any one prevents full marks.
- 2023S-Q31(a) requires first defining primary structure, then using a labelled diagram to explain peptide bond formation, 4 marks in total (also counted in Pattern 6).
- Common problems in Section Three listed in the 2018 report include "drawing peptide links"; when drawing, check each amide group.

Past exam questions: 2016S-Q36(ai), 2016S-Q36(aii), 2016S-Q36(b), 2017E-Q40(a), 2018E-Q39(a), 2018E-Q39(b), 2019E-Q41(a), 2019E-Q41(b), 2020E-Q35(e), 2021E-Q34(a), 2023S-Q31(a), 2024E-Q37(e), 2025E-Q36(d)

Pattern 6: Primary, secondary and tertiary structure of proteins ★

Identify: the question contains "primary / secondary / tertiary structure", "α-helix", "β-pleated sheet", "disulfide bridge", "predominant interaction", "substitution of an amino acid", "compare the sequences".

Steps

1. Primary structure: the sequence of amino acids, joined by peptide bonds. Write the definition as "the sequence of α-amino acids".
2. Secondary structure: hydrogen bonds form between N–H and C=O on the backbone, causing parts of the peptide chain to become:
    - α-helix: within the same chain, the two amide groups connected by hydrogen bonding are at least four apart (key wording), forming a helix;
    - β-pleated sheet: hydrogen bonds form between adjacent peptide chain segments, forming a pleated sheet. When drawing hydrogen bonds, use dashed lines to connect the oxygen of C=O (lone pair) to the hydrogen of N–H.
3. Tertiary structure: the three-dimensional shape into which the whole peptide chain folds, formed by interactions between side chains: disulfide bonds (Cys with Cys, covalent, strongest), hydrogen bonds, dipole-dipole forces, dispersion forces and ionic bonds. Definition statement: "the overall three-dimensional shape of the protein, formed by side-chain interactions".
4. Substitution or mutation: first identify the substituted R group, then determine which interaction changes. For example, with no cysteine there is no disulfide bond; the ring structure of proline can only form weaker dispersion forces, making the structure more open and more flexible.
5. Comparing two sequences: first count the identical amino acids at the same positions, then state one difference and its position.
6. Forces between side-chain pairs: determine from the nature of the side chains: Ala–Val involves dispersion forces, Gln–His involves hydrogen bonding, Cys–Cys involves a disulfide bond.

Marking points (typical key wording): definition of primary structure (2022S-Q29(c): amino acid sequence;

the key for 2023S-Q31(a) also requires stating the sequence and peptide bonds); the hydrogen bond of secondary structure is correctly identified as being between N–H and C=O (2017E-Q40(c), 2 marks; 2023S-Q31(b)); the key wording for α-helix is "hydrogen bonds in the same chain + spiral shape" (2017E-Q40(c)); definition and formation of tertiary structure, with the points "three-dimensional shape + side-chain interactions" (2022S-Q29(bii), 3 marks total: 1 mark for definition, 1–2 marks for formation); the three side-chain force pairs are 1 mark each (2020E-Q35(c): Ala–Val dispersion forces, Gln–His hydrogen bonding, Cys–Cys disulfide bond; dipole-dipole is also accepted); after mutation, side-chain interactions change and folding differs (2025E-Q36(e)); without cysteine there is no disulfide bond and the structure is more open (2025E-Q36(f)); 2025E-Q36(c): the helix forms hydrogen bonds between carbonyl oxygen and N–H, occurs within the same chain, and is at least four amide groups apart.

Common errors (exam reports)

- Describing the hydrogen bonds of secondary structure as forces between R groups. The key requires hydrogen bonds between N–H and C=O in amide groups (2023S-Q31(b)).
- Confusing peptide bonds of primary structure with disulfide bonds of tertiary structure. Primary structure is determined by peptide bonds; disulfide bonds are covalent bonds between side chains and belong to tertiary structure.
- 2024E-Q37(e) requires explaining the formation of a hydrogen bond; if you omit "the partially positively charged H and the lone pair on the oxygen of another part", you lose marks.
- 2024E-Q37(f): a disulfide bond can only form between cysteines; answering "cysteine" is sufficient.
- The key for 2019E-Q41(c) requires distinguishing primary and secondary structure: primary is the sequence, secondary is the α-helix or β-pleated sheet formed by hydrogen bonding.
- Report: in this note, the report entries for 2020E-Q35, 2024E-Q37 and 2025E-Q36 are blank, so reasons for lost marks cannot be added from the reports.

Past exam questions: 2016S-Q29(e), 2017E-Q40(b), 2017E-Q40(c), 2017E-Q40(e), 2018E-Q39(e), 2019E-Q41(c), 2020E-Q35(a), 2020E-Q35(b), 2020E-Q35(c), 2020E-Q35(d), 2021E-Q34(b), 2021E-Q34(c), 2022S-Q29(bi), 2022S-Q29(bii), 2022S-Q29(c), 2022S-Q29(d), 2023S-Q31(a), 2023S-Q31(b), 2024E-Q37(d), 2024E-Q37(e), 2024E-Q37(f), 2024E-Q37(g), 2025E-Q36(c), 2025E-Q36(e), 2025E-Q36(f)

Pattern 7: Enzymes as biological catalysts

Identify: the question asks "what type of organic compound is an enzyme", "role of the enzymes", "justify the conditions", "which catalyst is industrially preferred", "to which class of biological chemicals does lipase belong".

Steps

1. Classification: enzymes are proteins (polypeptides) that act as biological catalysts.
2. Catalytic action: enzymes lower the activation energy of a reaction and provide an alternative reaction pathway, increasing the proportion of effective collisions; they speed up the forward and reverse reactions equally and do not change the position of equilibrium.
3. Specificity: they catalyse only specific substrates.
4. Conditions: enzymes are only effective over a narrow pH and temperature range; a temperature that is too high or a pH that is too acidic or alkaline causes the enzyme to denature and lose activity.
5. Industrial comparison: compared with strong base (NaOH) catalysis, enzymes usually have lower energy requirements, do not produce soap, can be reused, work under milder conditions and are more environmentally friendly.

Marking points (typical key wording): enzymes are proteins or polypeptides (2016E-Q38(a)); the three points for the role of a catalyst are 1 mark each: increase reaction rate, provide an alternative pathway, lower activation energy (2025E-Q38(b)); the narrow temperature and pH range of enzymes, or loss of function (2023E-Q39(ai): justify the fermentation conditions; 2024E-Q37(c)); industrial advantages (2024E-Q37(c): lower energy consumption, fewer side reactions, easier purification, sustainability — any one); three reasons lipase is used as an industrial catalyst (2020E-Q40(f): avoids soap formation, low energy consumption, reusable, low temperature, low toxicity — any three); enzymes belong to the enzyme class (2020E-Q40(b)).

Common errors (exam reports)

- Saying a catalyst can increase yield. The key states clearly that a catalyst speeds up the forward and reverse reactions equally, so yield does not change (the key for 2021E-Q39(e) states that lipase does not affect yield).
- In 2016S-Q28(b), when comparing the two catalysis methods, you must clearly state which method you are referring to (“lipase-catalysed” or “base-catalysed”), and each point must be a comparison.
- Stating only that “enzymes are catalysts” without explaining activation energy and pathway will not earn full marks.
- The 2024 report warns: do not use anthropomorphic words such as “sensitive” to describe enzymes (e.g. “enzymes sensitive to pH”); instead write denaturation or outside the optimum range.

Past exam questions: 2016E-Q38(a), 2016S-Q28(b), 2020E-Q40(b), 2020E-Q40(f), 2021E-Q39(e), 2023E-Q39(ai), 2024E-Q37(c), 2025E-Q38(b)